

Work Sheet 4 (Data Structures)

1. Problem Explanation: Printer Task Scheduling

Imagine a printer that can only print one document at a time. Documents are processed based on their priority. Each document in the queue has a specific priority level, with higher priority numbers indicating higher priority.

Documents are stored in a queue as they arrive. The printer checks the document at the front of the queue: if it has the highest priority among all documents in the queue, it is printed. If not, it is moved to the back of the queue.

Input and Output Specifications

1. Input:

- n : an integer representing the number of documents in the queue.
- priorities : a list of n integers, each representing the priority of a document.
- m : an integer representing the position (0-indexed) of the document of interest.

2. Output:

- An integer representing the number of operations required for the printer to print the document at position m .

Example Input:

$n = 5$

$\text{priorities} = [2, 1, 3, 2, 5]$

$m = 2$

Example Output:

5

Explanation:

- The document at position 2 (priority 3) will be printed after 5 operations.

Tip: To solve this problem, we use a queue data structure. Here's the step-by-step approach:

1. Initialize the Queue:

- Store each document in the queue with its priority and index.

2. Process the Queue:

- Check if the document at the front has the highest priority.
- If it does, print it and increment the count.
- If not, move it to the back of the queue.

3. Continue Processing:

- Repeat until the document at position m is printed, then return the count.

Note: Use Queue ADT as discussed in class.

2. Traffic Light Queue Management

A traffic intersection has a single traffic light, which allows vehicles to pass through the intersection one at a time. Vehicles arrive at the intersection with a **wait time** (time they must wait at the red light) and are released one by one when the light turns green.

Given a list of vehicles arriving at the intersection and the **green light duration** (time each green light cycle lasts), calculate the total **wait time** each vehicle experiences from its arrival until it is allowed to pass through the intersection.

Input:

1. An integer n , representing the number of vehicles arriving at the intersection.
2. A list of n integers where each integer represents the time (in seconds) each vehicle requires to cross the intersection.
3. An integer `green_light_duration`, representing the total time (in seconds) the light stays green in each cycle.

Output:

- A list of integers where each integer represents the **total wait time** for each vehicle from its arrival to when it crosses the intersection.

Example:

Input: n = 5

vehicle_times = [2, 4, 1, 3, 5]

green_light_duration = 6

Output: [0, 0, 6, 9, 12]

Input: n = 6

vehicle_times = [3, 5, 2, 6, 1, 4]

green_light_duration = 7

Output: [0, 3, 7, 13, 14, 20]

Draw a queue processing steps for each example.

- 3. Open link and solve. A bonus marks of 2 can be judged in quiz for understanding problem. And 3 marks will be awarded on complete solution.**

Note: Copy and paste URL in browser, instead of clicking it.

https://onlinejudge.org/index.php?option=onlinejudge&Itemid=8&page=show_problem&problem=3252